SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

The OptiCrop Smart Agricultural Production Optimization System uses HTML, CSS, and JavaScript for the frontend, Python with Flask for the backend, MySQL for data storage, Scikit-learn for machine learning, and GitHub for version control. This technology stack ensures efficient crop prediction, secure data management, and easy maintenance.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, JavaScript	Creates a responsive and user-friendly interface.

2	**Backend / Server-Side**	Python, Flask	Processes user input and integrates the ML model.
3	**Database / Data Storage**	MySQL	Stores user, soil, crop, and prediction data securely.
4	**Cloud / Hosting / Deployment**	Render / PythonAnywhere (or Localhost if not deployed)	Hosts the web application for online access.
5	**Version Control & CI/CD**	Git & GitHub	Manages source code and enables collaboration.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib	Used for machine learning, data preprocessing, analysis, and visualization.